INTRODUCTION TO ICTs

Duration: 28 Periods

Competency: The learner operates a variety of ICTs to perform tasks in day-to-day life.

Learning Outcomes:

a) explore the utilization of various ICT tools in day-to-day life. (k, u, a, gs)
b) use digital tools to solve day-to-day life challenges. (k, s, v, gs)
c) create directories and use them to manage electronic files. (k, u, s)

ICT stands for Information and Communication Technology.

Definition of ICT

ICT refers to the diverse set of technological tools and resources that are used to create, share, communicate, disseminate, store and manage information by electronic means.

6-Action Rule (Definition Break Down)

Action	**Technical Meaning**	**Real-World Example**
Create	Generating new content digitally.	Typing a report or recording a video.
Manage	Organizing, updating, and controlling access to data.	Organizing files into folders or updating a student database.
Share	One-to-one or small group exchange.	Sending a file via Bluetooth or Email.
Communicate	Two-way interaction.	A WhatsApp call or a Zoom meeting.
Disseminate	Spreading information to a mass audience.	Posting on a school website or a Radio broadcast.
Store	Saving data for future use.	Saving work on a Flash Disk or in the Cloud.

Information

Information refers to processed or organized data that is meaningful and useful to the person receiving it. It provides context, relevance, and purpose, allowing people to make decisions, solve problems, or understand situations better.

Data refers to the raw facts, figures, numbers and symbols that are entered into a computer for processing and carries less meaning to the user or the person receiving it. Data on its own (like raw numbers) doesn't tell a story, but *information* provides a message.

Examples of information

- A student's Report Card (processed from raw marks).
- A Weather Forecast (processed from temperature and wind readings).
- A Timetable (organized data about time and subjects).

Communication

Communication in ICT is the process of sending, receiving, and exchanging information between two or more parties using electronic devices and media. It involves a sender, a message, a medium (channel), and a receiver.

Examples:

- Sending a Text Message: A one-to-one exchange.
- A Radio Broadcast: One-to-many dissemination of news.
- A Telephone Call: Real-time, two-way verbal exchange.

Situation where communication takes place

- A teacher sends an email to students with homework instructions.
- A student attends an online class using Zoom or Google Meet.
- A school sends SMS messages to parents about upcoming events.
- Students chat in a WhatsApp group to discuss a group project.
- A head teacher uses a school management system to share exam results with parents.

Technology

Technology refers to the application of scientific knowledge, tools, methods and skills to solve problems or perform specific tasks. It is the "how" we get things done more efficiently and easily.

Examples:

- Hardware: A computer, a projector, or a smartphone.
- Software: An operating system (Windows/Android) or an App.
- Infrastructure: Fiber optic cables or satellite dishes.

Different ICT tools and their uses

Hardware

Computer (Desktop or Laptop)
A powerful digital device used to input, process, store, and display information. It runs various software applications for tasks such as word processing, spreadsheet management, programming, internet browsing, and graphic design.

Smartphone
A portable device that combines a mobile phone with advanced computing functions. It allows users to make calls, send texts, access the internet, take photos, use apps, and perform various communication and productivity tasks.

Tablet
A lightweight, touchscreen-based device larger than a smartphone but smaller than a laptop. It is used for reading digital books, accessing educational content, taking notes, browsing, and media consumption.

Printer
An output device that produces hard copies of digital documents, images, or graphics on paper. Commonly used in schools, offices, and homes for printing assignments, reports, or photos.

Scanner
An input device that captures images or text from physical documents and converts them into digital format for editing, emailing, or storing on a computer.
Projector
A display device that projects images, videos, or presentations from a computer or media player onto a larger screen or wall, often used in classrooms and meetings for group viewing.
Digital Camera / Webcam
A device used to capture still images or video. Digital cameras are used for photography, while webcams are mostly used for live video communication in online meetings or virtual classrooms.
Photocopier
A machine that quickly makes duplicate copies of documents or images, widely used in offices and schools to reproduce handouts or reports.
Flash Drive (USB Drive)
A small, portable device used for storing and transferring digital files between computers and other devices. It is convenient for backing up data or carrying work between home and school.
External Hard Drive
A large-capacity storage device used to back up or transfer large amounts of data such as videos, software, and databases. It connects to a computer via a USB or wireless connection.
Modem
A device that connects a home, office, or school to the internet by converting signals from your Internet Service Provider (ISP) into digital form that devices can use.
Router
A device that distributes internet connection from the modem to multiple devices—either through Wi-Fi or Ethernet cables—allowing phones, computers, and other gadgets to access the internet at the same time.
Interactive Whiteboard (Smartboard)
A touch-sensitive board that connects to a computer and projector, allowing users to write, draw, and interact with digital content using a pen or finger. It enhances teaching and learning through interactive lessons.
Headphones / Earphones
Audio output devices worn over or in the ears to listen to sound privately from a computer, phone, or other digital device. Useful for online learning, editing audio, or quiet environments.
Microphone
An input device that captures voice or sound and converts it into digital signals. It is used for voice recording, video conferencing, dictation, and virtual communication.
Speakers
Output devices that convert digital audio signals into sound. Used for playing music, videos, or audio instructions during lessons or presentations.
Photographic Drone
A remote-controlled flying device equipped with a camera, used to capture aerial images and videos. Common in media, surveying, security, and agriculture.
UPS (Uninterruptible Power Supply)
A backup power device that provides temporary electricity to a computer or other ICT equipment when there is a power cut. It prevents data loss and gives users time to save their work and shut down safely.
Barcode Scanner
A device used to read barcode labels, commonly found in libraries, shops, and warehouses for tracking items and speeding up transactions.
Surge Protector (Power Extension with Surge Control)
A device used to protect ICT equipment from damage caused by electrical surges or spikes.

Network Switch / Hub
Hardware used in offices and schools to connect multiple computers or devices within a local network so they can share files and resources like printers.

Software

Video Conferencing Tools (e.g., Zoom, Microsoft Teams)
Software platforms that enable real-time video and audio meetings over the internet, commonly used for online classes, virtual meetings, webinars, and remote collaboration.
Email Applications (e.g., Gmail, Outlook)
Digital tools that allow users to send, receive, and organize messages and files electronically, often used for communication in academic and professional environments.
Web Browsers (e.g., Google Chrome, Mozilla Firefox)
Software that allows users to access and navigate the World Wide Web, view websites, search for information, and use online services.
Learning Management Systems (e.g., Google Classroom, Moodle)
Platforms used by schools and organizations to manage digital learning. They support uploading of materials, giving assignments, grading, and communication between teachers and learners.
Antivirus Software
A program installed on computers and mobile devices to detect, block, and remove harmful software (viruses, malware, spyware). It protects systems from being damaged or hacked.
Firewall
A security tool (either software or hardware) that monitors and controls network traffic. It helps prevent unauthorized access to or from a private network.
Cloud Storage Tools (e.g., Google Drive, Dropbox)
Online platforms that allow users to store files safely on the internet, access them from any device, and share them with others.
Software Applications (e.g., Microsoft Office, Adobe Reader)
Programs that help users perform specific tasks such as typing (Word), creating spreadsheets (Excel), designing posters (Publisher), or reading PDF files (Adobe Reader).
Backup Software
Used to create copies of important files and system settings, which can be restored in case of data loss or system failure.
Point of Sale (POS) Software
A system used in shops and businesses to manage sales transactions. It records what is sold, calculates the total price, prints receipts, and can track stock levels and generate sales reports.
Accounting Software (e.g., QuickBooks, Tally)
Used to manage financial records such as income, expenses, payroll, and invoices. Helps with budgeting, tax calculations, and financial reporting.
Word Processing Software (e.g., Microsoft Word, Google Docs)
Allows users to type, edit, format, and print text documents such as letters, reports, and assignments.
Spreadsheet Software (e.g., Microsoft Excel, Google Sheets)
Used to organize and analyze data using rows, columns, and formulas. Commonly used for budgeting, record-keeping, and calculations.
Presentation Software (e.g., Microsoft PowerPoint, Google Slides)
Helps users create slideshows with text, images, animations, and videos—useful for teaching, business presentations, and student projects.
Database Software (e.g., Microsoft Access, MySQL)
Stores large amounts of organized data for easy retrieval and updating. Used in schools, hospitals, and banks to manage records.

Graphics and Design Software (e.g., Adobe Photoshop, Canva)
Used for creating and editing images, posters, logos, and other visual content. Useful in marketing, education, and creative industries.
Web Design Software (e.g., WordPress, Dreamweaver)
Helps users create and manage websites without needing deep programming skills.
Learning Platforms (e.g., Moodle, Google Classroom)
Allow teachers to deliver lessons, post assignments, grade student work, and communicate with learners online.
Internet Browsers (e.g., Chrome, Firefox, Safari)
Software used to search for and view websites, download files, and access online tools.
Media Players (e.g., VLC, Windows Media Player)
Play audio and video files, useful in both entertainment and education for listening to music, podcasts, or viewing learning videos.
PDF Readers (e.g., Adobe Acrobat Reader)
Software used to view, print, and sometimes edit PDF documents—commonly used for forms, e-books, and reports.
Communication Software (e.g., WhatsApp, Zoom, Microsoft Teams)
Used for sending messages, making video/audio calls, and conducting virtual meetings or lessons.

A Computer and its Parts

A computer is an electronic device which inputs, processes and stores data or information and outputs information when given a set of instructions. Computer instructions are known as commands.

A Computer is an electronic device that accepts data input, processes it according to some specified instructions, outputs the information and stores the results for future use.

The operating speed of a computer is in Millions of Instructions per second **(MIPS).**

What a computer does?

There are four functions that a computer is built to perform

- **Input:** Computers receive data and instructions through input devices like keyboards, mice, touch screens, etc.
- **Processing:** The computer's central processing unit (CPU) interprets and executes instructions, performing calculations and manipulating data according to the program's logic.
- **Storage:** Computers store data and programs in various types of memory, such as RAM (random-access memory) for temporary storage and hard drives or solid-state drives for long-term storage.
- **Output:** After processing the data, the computer produces output through output devices like monitors, printers, speakers, etc.

Characteristics of modern computers

Speed: Computers are quite fast in their operation in that their speed is measured in Millions of Instructions Per Second (MIPS).

Diligence : This refers to the computers' ability to perform the same task over and over (repeatedly) for a long time without getting bored or tired.

Accuracy: This is the computer's ability to process large amounts of data without making mistakes. They can process large amounts of data and generate error- free results, provided the data entered is correct. It follows therefore, that if wrong data is fed into the computer, the resulting output will be incorrect. Hence the saying; Garbage In Garbage Out (GIGO).

Automation: This is the ability of computers to carry out tasks without instruction by the user. Such tasks include; automatic detection and cleaning of viruses. This is at times because of the instructions (programs) installed in them.

Storage: This is the ability of computers to keep/ retain data or information either temporarily or permanently.

Artificial intelligence: This is the computers' ability to mimic human behavior.

Versatility : This means that a computer has the ability to perform more than one task at the same time.

Communication. They can communicate with other computers to share data, information, programs and instructions through networks. Networked computers have the ability to communicate and share resources.

Basic Parts of A computer

The basic parts of a computer include the following.

The system unit

It's rectangular box that houses the internal electronic delicate parts of a computer.

The most important part is the CPU or processor which act as the brain of a computer.

Mouse

A mouse is a small device used to point and select items on the computer screen. It used in the movement of the cursor

Keyboard

A keyboard is in input device mainly used for typing text and issuing commands into the computer.

A monitor is an output device that displays information in visual form like text and graphics to the user.

A speaker is an output device that is used to play sound. They may be built into the system unit or connected with cables.

Microphone - An input device that captures sound, allowing users to record voice, speak in meetings, or give voice commands.

Computer System

A computer system is a collection of computer components working together to input data, process data and output results called information.

Computer components may include hardware, software, people (users), procedures and information.

Components of a computer system

- Hardware
- Software
- Computer users/Human ware/Liveware
- Procedures
- Information

Computer Hardware

Hardware refers to the physical components of a computer system. Any component you can touch, feel and pick up is called a computer hardware

Examples of hardware computer components include; Keyboard, mouse, monitor, printer, hard drive etc.

Computer hardware includes;-

- Input devices like the keyboard and mouse,
- Processing devices like the Microprocessor Chip,
- Storage devices like the Hard disks and the CDs,
- Output devices like the monitor and the printer
- Communication devices like Modem, Router, Switch, NIC, VGA Cable, HDMI Cable etc.

Computer software

Software refers to the set of electronic instructions that tell the computer hardware what to do and how to do it.

Software is a term for electronic instructions that tell the computer how to perform a task.

These are a series of programs (instructions) that tell the computer what and how to work. Computer software can be grouped into System software and Application software

Information

Information refers to processed data that has meaning and value to the end user. When data is processed, it gives meaning. For example, a receipt gives information about purchased items and a report card gives information about a student's performance.

Procedure

A procedure refers to a sequence of actions conducted in a certain order to accomplish a given task or make sure that a computer system run smoothly.

For example, storing data on a flash disk or external hard disk requires a series of procedures like inserting the disk into the computer, finding the information to be stored, copying it and putting it on to the disk and finally removing the disk from the computer

Users/Human ware/Liveware

Computer Users (Human ware) refers to the people who operate and initialize instructions to the computer system.

They design and develop computer systems, operate the computer hardware, create the software, and establish procedures for carrying out tasks.

Types of computer users

Ordinary user -is someone without much technical knowledge of computers but uses computers to produce information for professional or personal tasks, enhance learning, or have fun. Ordinary users include Computer students, Typists (Secretaries), etc.

Professional user -is a person in a profession involving computers who has had formal education in the technical aspects of computers; Examples include Computer programmers, webmasters, etc.

Activity: Classifying Components of the computer system

1. Classify the following computer components under the categories: hardware, software, information, procedures and people

Computer technician, flash disk, Windows 10, S.5 ICT Exam, Secretary, DVD Drive, Microsoft Word, ICT user manual, creating and printing a letter, photocopier

2. Imagine you have been invited by your local council chairperson to educate people in their community about hardware and software. How would you explain the difference between hardware and software?

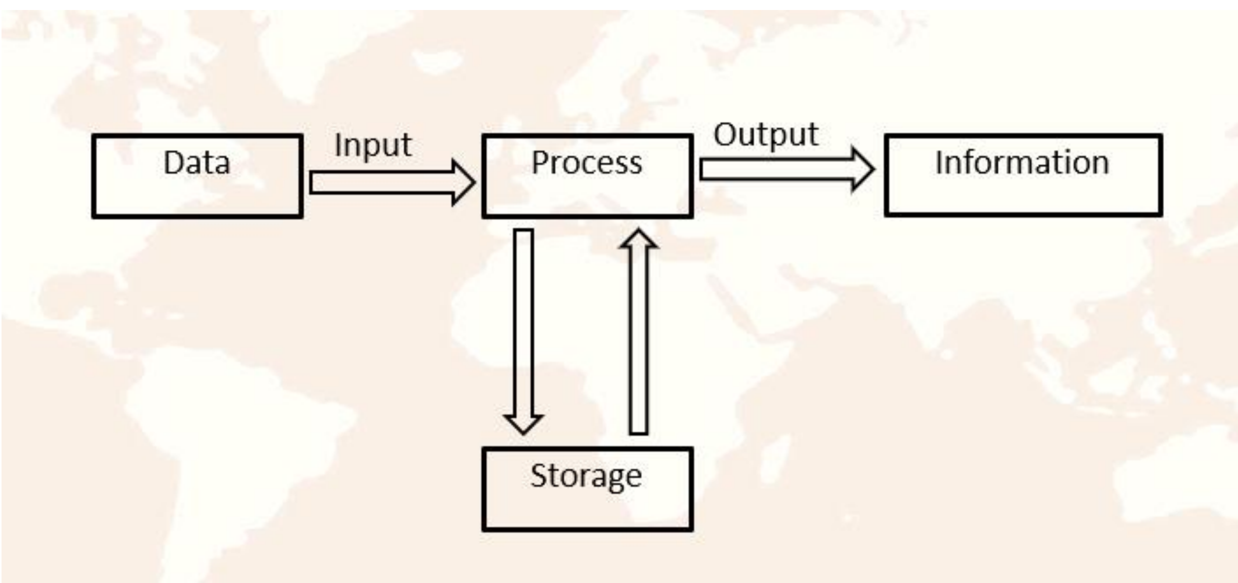
The Information Processing Cycle

The information processing cycle is a series of steps the computer follows to receive data and process it into information.
The 4 basic operations of the information processing cycle are input, processing, output and storage.

Data processing

Data processing is the activity of manipulation of raw facts to get meaningful information.
These are the steps that are taken to convert raw facts, which are data, into information.

Information Processing Cycle



Stages of Information processing cycle

Input stage

Data, instructions and commands are entered into the computer with the help of input hardware devices for processing.

Processing stage

Data is converted, changed, transformed, interpreted, manipulated into information that can be understood by the user.

Output stage.

At this stage, information is produced, conveyed or displayed to the user. The user can be able to see the results of processing with the help of output devices.

Storage stage.

Information is stored temporarily during processing or permanently for future use in the computer.

USES OF ICTs IN SOCIETY

Area where ICT tools are used

- Transport
- Education,
- Business
- Health
- security
- Politics and Governance
- Communication
- Entertainment / leisure
- Technical and scientific uses

How ICTs are applied in Transportation

- GPS and Navigation Systems: ICT tools like GPS devices and mobile apps (e.g., Google Maps) help drivers and passengers find the best routes, avoid traffic, and reach destinations faster.
- Traffic Management Systems: Traffic lights, road sensors, and surveillance cameras use ICT to monitor and control traffic flow in busy areas, helping to reduce congestion and accidents.
- Electronic Ticketing and Payments: Buses, trains, and airlines use ICT systems for booking tickets online, checking in, and making cashless payments using cards or mobile money.
- Fleet Management: Transport companies use ICT tools to track their vehicles, monitor fuel use, plan deliveries, and communicate with drivers in real time.
- Public Transport Information Systems: Passengers can access information about bus or train schedules, arrival times, and delays through websites, mobile apps, or digital signboards.
- Security and Surveillance: CCTV cameras and tracking systems are installed in stations, airports, and vehicles to monitor activities and ensure safety.
- Vehicle Safety Features: Modern vehicles use ICT-based systems like automatic braking, lane assist, and collision alerts to reduce the risk of accidents.

Application of ICTs In the Area of Education

- Computers can be used to help students learn a new subject on their own through Computer Aided Instructions (CAI)
- Students use computer software to learn at their own pace (speed) and answer questions through Computer Assisted Learning (CAL)
- Teachers can use computers to mark and assess the students' performance through Computer Assisted Assessment (CAA). This reduces the time and labor spent marking the students' scripts.
- Schools use computers to create school websites for sharing information with the public.
- Productivity tools like desktop publishing and presentation software are used in projects and other school activities.
- Computers are used for calculating mathematical arithmetic by students and teachers in class.
- With Use of School Administration and Management Systems. (SAMS) Records management is made easier because all details of learners can be held on computer, and easily retrieved, reducing administrative costs.
- Students' Progressive Report Cards can be produced electronically by use of computers instead of hand written ones.
- Distance learning through computer-based training. People get award such as degrees without going to class.
- Teachers use simulation software to perform difficult or dangerous experiments in class.
- Use of special facilities for students with disabilities like text to speech and speech recognition to help blind students.

Application of ICTs In The Area Of Business

- Computers enable people to Work from home, using a computer connected to the employer's network or via the Internet. This is known as **Telecommuting**.
- Computers have created more jobs such as Computer technicians, Computer teachers, etc.
- Buying and selling Computers and its components is a source of income to individuals, and companies.
- Through, Computer Aided Design (CAD), scale drawings, and excellent designs can be created easily.
- Computers are used for sending and receiving Mobile Money and making worldwide money transfers.
- Banks use Computers to manage transactions and Automated Teller Machines ATMs for 24 hours banking.
- Computers help in Business Advertisement through creating websites, internet, flyers, brochures and billboards.
- Computers are used in typesetting business for production of document printouts and publication of Books for sale.
- Computers are used for E-Commerce: the sale of goods and services over the internet.

Applications of ICTs in the Area of Health

- Hospitals use computers for managing and storing Records electronically, rather than paper files.
- Hospital Administration is also aided by printing labels, allocating beds, make appointments, staff rotas, etc.
- Internet helps us get Web sites for information on health care, treatments, conditions, etc.
- Monitoring/Diagnosis such as Heart rate, blood pressure, etc. is aided by Computer Expert systems.
- Medical Training is facilitated by Simulation software and on-line data sources.
- Computers are used to carry out many surgical procedures such as laparoscopic surgeries
- They are used in diagnosis and cure of many disease for example CT Scan, Ultra sound devices and Magnetic imaging.(MRI)
- They enable online consultations by medical professionals.
- Use of computer assisted tests can be carried out before prescribing treatment.
- Use of computer assisted life saver machine.
- Enable faster communication between patients and doctors
- Can be used to monitor the patients in hospitals e.g. CCTVs

Application of ICTs in the Area of Security

- Computers aid monitoring security through cameras, Automatic number plate recognition, etc.
- Communication systems are widely used in the military to coordinate the personnel.

- Some computer systems can detect temperatures and alarm in case of danger of fire outbreaks.
- Computers are used for capturing data for Police National Computer Databases –, vehicle number plates, criminal's fingerprints, etc.
- Computers are used to detect presence of illegal devices such as bombs.
- Computers are also used for controlling dangerous weapons such as missiles.
- Computers are used for storing criminal databases and information at Police stations.

Uses of ICTs in the Area of Politics and Governance

- Paying government tax online through a government website.
- Online forms such as vehicle registration and passport forms.
- Advertising government tenders and Applying for government tenders
- Public records -A maintained database of statistical information such as electoral register and census data can be availed online.
- Use of electronic voting during elections.
- Government departments can use a computer-based platform to get feedback from the citizens.

Uses of ICT in Art, Leisure and Entertainment

- Computers enables people to play computer games like GTA Vice City, Need for Speed, Solitaire etc.
- Internet has promoted social networking that has enabled interaction between people. Examples include Twitter, Facebook, LinkedIn etc.
- Computers can be used to play music during free time.
- Computers can be used to watch movies and videos.
- Digital cameras can be used to record and capture videos on parties.
- Computers are used to compose and edit songs by producers through using audio and video production software.
- Internet enables people to read magazines online.

Uses of ICTs in the Area of Communication

- E-mail: Electronic Mail sent from one person to another using connected computers helps a lot in the area of communication.
- Video Conferencing enables people in different locations to conduct meeting as if they are in the same location.
- Computers are used for Faxing: Sending an image of a document electronically.
- Computers enable people to send voice, image, text and data though telephones and mobile cell phones.
- Social Networks such as Facebook, and Twitter enable people to stay in touch with their relatives, friends and interests.

Technical and Scientific Uses of ICTs

- In Astronomy, Computers are essential tools to study the behavior of the complex systems in space as regards to their movements, interactions etc.

- Through Computer Aided Manufacture (CAM), computers can be used to control the production of goods in factories.
- Monitoring highway traffic

- Computers are used to tell schedules of water vessels, train, buses to their respective stations.
- Computers are used very extensively in design of roads. Roadways and bridges are designed using software programs like CAD etc.

Application of ICTs in Security

- One major application is in surveillance systems, where Closed-Circuit Television (CCTV) cameras are used to monitor and record activities in real-time, helping to prevent crime and gather evidence.
- ICT is also used in access control systems, such as biometric scanners that use fingerprints, facial recognition, or ID cards to restrict access to sensitive areas like offices, banks, or data centers.
- Alarm systems powered by ICT detect unauthorized entry or unusual movement and alert security personnel.
- Cybersecurity tools like firewalls, antivirus software, and encryption help protect computer systems and digital information from hackers, viruses, and data breaches.
- ICT also enables quick and effective communication among security teams through mobile phones, radios, and walkie-talkies, especially during emergencies.
- GPS tracking systems are used to monitor the movement of vehicles or people, which is particularly useful in law enforcement or logistics.
- ICT supports data storage and retrieval, allowing authorities to maintain databases of criminal records, fingerprints, or surveillance footage for investigation and legal purposes.

Computer System Assembly and Start-up

Assembling means to put together different components of something.

Computer system assembly refers to the process of physically putting together the individual components of a computer.

While technical people can assemble all the parts inside the system unit, you will use a pre-assembled system unit and connect it with other parts of the computer.

It is important to understand the importance of each part of the computer that you will connect and the port where it is connected. Port is a technical term that means an opening on the system unit where you can connect a peripheral device to work with the computer.

Some of the parts of the computer you may need to assemble to enable the computer to function include the following:-

- Mouse
- Keyboard
- Monitor
- Power cables
- System unit and
- UPS (Uninterruptible Power Supply)

Booting
The process of starting a computer and loading an operating system into memory to make the computer ready for use.

Types of Booting
Cold Booting
Cold booting refers to the process of starting a computer from a completely powered-off state. It occurs when the system is turned on after being shut down, where the power supply is activated and the hardware components undergo initialization.

Warm Booting
Warm booting refers to the process of restarting a computer **which has been running or operating.** This typically occurs when the operating system is restarted using a command, such as selecting "Restart" from the operating system's start menu or pressing the **Ctrl + Alt + Delete** keys in Windows.

Reasons for warm booting

- Before and after installation of a new application or utility software
- After uninstalling an application or utility software
- When a peripheral, hardware component fails to function
- After uninstalling or disconnecting a hardware component
- During and after installation of operating system
- During installation of a new computer hardware
- After configuring a server on a network or changing network settings
- After changing system control settings
- When a user wants to clear a malicious infection like malware, spyware, viruses from the memory
- When switching from one operating system to another – multiple OS system
- When the computer stops responding to commands (freezes or deadlocks)
- After malware or virus scanning
- When there is suspected system attack or tapping
- When the computer system slows down
- When application software fails to load or work
- When power voltage suddenly lowers

Benefits of restarting a computer

- It flushes RAM and removes other files that could be causing computer freezing.
- Fast Performance. Reboots are known to keep computers running quickly.
- Stops Memory Leaks. These occur when a program doesn't close properly.
- Fixes Internet Connection.
- Recovers the computer from errors

Step-by-Step Process of Booting
- When the computer is powered on, the power supply activates and sends an electrical signal to the motherboard and other connected devices, such as the RAM, hard drive, and peripherals, preparing the system for booting.
- The CPU resets to prepare for the boot process. It then looks for the ROM (Read-Only Memory) where the BIOS (Basic Input/Output System) is stored, as it manages hardware initialization.

- The BIOS performs the Power-On Self-Test (POST), a diagnostic routine that checks essential hardware components such as RAM, CPU, keyboard, storage devices, and graphics card to ensure they are properly connected and functioning.
- The POST results are compared with data stored in the CMOS (Complementary Metal-Oxide-Semiconductor) chip, which holds the system configuration and settings. Any discrepancies or errors will result in an error message or beep codes to alert the user.
- If the hardware checks pass, the BIOS searches for the bootloader program, which is typically located on the primary storage device (like an HDD or SSD).
- The bootloader is loaded into memory and executed. It then loads the kernel of the operating system into RAM. The kernel is the core component that manages hardware resources and enables software to interact with the system.
- Once the kernel is loaded, the operating system continues to load configuration files, device drivers, and other system components into RAM.
- Once everything is initialized, the desktop and icons appear on the screen, indicating that the computer is ready for use.

Steps to assemble a working computer given system unit, flat screen monitor, keyboard, mouse, VGA cable, power extension cable, UPS, Printer, printer data cable and 3 power cables.

- Connect the mouse to the PS/2 mouse port or USB port
- Connect the keyboard to the PS/2 keyboard port or USB port
- Connect the printer to the ISB port using the available printer data cable.
- Connect the first power cable to the power port on the printer to draw power from the power extension cable.
- Connect the second power cable to the power port behind the system unit to draw power from the UPS.
- Connect the third power cable to the power port at the base of the flat monitor to draw power from the UPS.
- Connect the VGA cable from the monitor to the video port behind the system unit
- Connect the UPS to draw power from the extension cable.
- Connect the power extension to the wall socket
- You can finally switch on the power source for the extension cable

Procedures to start a computer properly

- Switch on the wall socket
- Switch on the power extension cable, power stabilizer and the UPS in the same order.
- Switch on the monitor
- Switch on the printer
- Lastly switch on the system unit

Steps to Shut down a computer properly in Windows 10

- First close all programs that are running (close active windows)
- Click the Windows button (or Start button) to get system menu
- On the system menu, click power button
- Select Shut Down Command. The computer will display the screen preparing to shut down. The computer finally, turn off the system unit

- Switch off the monitor
- Switch off the power source (I.e., UPS, power extension and wall socket)

Activity: Make a write-up about the significance of observing proper start-up and shutdown of ICT devices.

FILE AND FOLDER MANAGEMENT

Files

A file is a collection of related data or information stored together on a computer under a specific name called file name.

A file is a basic unit of storage in a computer identified by a unique name.

Operation performed on a file

Computer users and programs can perform several basic operations on files. These operations allow for effective management, storage, and use of data. File operation includes

- Creating files
- Moving(Dragging from one location to another or cut)
- Copying
- Selecting single or multiple files (ctrl + click / Shift + click)
- Renaming
- Searching and Sorting by name, date or file extension (type)
- Printing
- Deleting
- Restoring

How to create a new file

Depending on the programs installed in your computer, you can create different types of files such as drawings, text document etc.

To create a new text document:

- On the free space on the desktop, right click.
- Point to new, click Text Document from the list available on application.
- Type a new name for the new file to replace the temporary name and press Enter key. NB: In windows, file name can contain up to 255 characters, including spaces but, with no special symbols such as \|/:*?”<>.

Selecting a Single File

Using the Mouse, Click once on the file you want to select. It will be highlighted to show it is selected.

Selecting Multiple Files

A. Selecting Files That Are Next to Each Other (Contiguous Selection)

Using the Mouse:

1. Click the first file.
2. Hold down the **Shift key**.
3. Click the last file in the group.
4. All files between the first and last will be selected.

Using the Keyboard:

1. Select the first file using arrow keys.
2. Hold down **Shift** and use arrow keys to extend the selection.

B. Selecting Files That Are Not Next to Each Other (Non-Contiguous Selection)

Using the Mouse:

1. Hold down the **Ctrl key**.
2. Click each file you want to select, one by one.
3. Release **Ctrl** when done.

Using the Keyboard:
Use Ctrl + Arrow keys to move, and Ctrl + Spacebar to select files one by one.

C. Selecting All Files in a Folder

Keyboard Shortcut:
Press Ctrl + A to select all files in the current folder.

Sorting files

- Right-click any open space within Windows Explorer and select Sort By.
- Choose to sort by Name, Date modified, Type, or Size.
- To view more sorting options, click More

Copying files

Cut or copy command are used to move or create a duplicate of an item respectively. When you cut or copy an item, it is temporarily held in a temporary storage location known as the clipboard.

Folders

A Folder is a virtual container for storing files and subfolders in the computer.
A folder is a named storage location where related files can be stored.
A folder is also known as a **directory** in some operating systems and has the following **features / properties:**

- A folder has a name
- A folder has a path originating from a special directory called **root directory**
- A folder has special access permissions for authenticated users.
- A folder or directory may be created inside another folder or directory. Such a folder or directory is called a subfolder or a subdirectory.

Creating a new folder

To create a new folder:

- Using my computer icon, on the folder tree on the left pane, select the location (desktop) in which you want to create a new folder.
- On the Home menu tab, Click new folder
- Type a new name for the folder to Replace the temporary name, then press Enter key.

Alternatively

- Right Click on the free space on the desktop,
- Point to New, click folder. Type a new name for the folder and press Enter

To copy a file or folder using File Explorer:

- Using my computer icon, display the Explorer window.
- Select the file or folder to be copied.
- On the Home menu, click copy.
- Select the drive or folder where you want the item to be copied.
- From the Home menu, click paste.
- Information or item is pasted to a new location.

Alternative Method (Using Right-Click):

- Right-click the file or folder → choose Copy.
- Go to the destination → Right-click an empty area → choose Paste.

Deleting Files and Folders

In Windows, when you delete an item from the hard disk, it is temporarily held in a special folder called the Recycle bin from where it can be restored if necessary.
Warning: Items deleted from a removable storage are not held in the recycle bin and are completely lost.

Steps to delete a file or folder

- Using This PC or File Explorer icon, display the Explorer window.
- Select the file or folder to be deleted.
- Right-click on the selected item.

- From the pop-up menu, click Delete.
- If prompted, confirm the deletion by clicking Yes
- The item is moved to the Recycle Bin.

Renaming files and Folders

Renaming refers to changing the previous name to a new name.

Steps taken to rename a file or folder

- Using **This PC** or **File Explorer** icon, display the Explorer window.
- Select the file or folder to be renamed.
- Right-click on the selected item.
  From the pop-up menu, click **Rename**.
- Type the new name for the file or folder.
- Press **Enter** to apply the new name.

Restoring deleted files and folders

- Double-click the **Recycle Bin** icon on the desktop to open it.
- Browse and select the file or folder you want to restore.
- Right-click on the selected item.
  From the pop-up menu, click **Restore**.
- The file or folder is restored to its original location.

Emptying the recycle bin

- Right-click on the Recycle Bin icon on your desktop.
- Select Empty Recycle Bin from the context menu.
- A prompt will appear asking if you are sure you want to permanently delete the items. Click Yes to confirm.

Moving a folder

To move a folder to a desired location:

- Click the folder and hold
- Drag and drop to the desired location

Alternatively

- Select the file or folder you want to move.
- Right-click on the selected item.
- From the pop-up menu, click Cut.
- Navigate to the destination folder or location where you want to move the file or folder.
- Right-click in the destination folder or location.
- From the pop-up menu, click Paste.
- The file or folder is now moved to the new location.

File extension

A file extension is a suffix at the end of a filename, typically consisting of three or four characters, that indicates the type of file. The extension helps the operating system understand which program to use to open the file.

Examples of file extensions

- **.txt** – Plain Text file, for notepad
- **.jpg** or **.jpeg** – Image file (JPEG format)
- **.doc –** Microsoft Word document (97-2003)
- **.docx** – Microsoft Word document (2007 and above)
- .**xls** – Microsoft Excel (97-2003) spreadsheet
- **.xlsx** – Microsoft Excel (2007 and above) spreadsheet
- **.pdf** – Portable Document Format (Adobe Acrobat)
- **.mp3** – Audio file (MP3 format)
- **.exe** – Executable file (Windows program)
- **.html** – Web page file (HTML format)
- .**mp4** – Video file (MP4 Format)

- **.exe** – Executable file
- **.ppt** - Microsoft Power point (97-2003) presentation
- **.pptx** – Microsoft Power point (2007 and above) presentation
- **.rtf** – Rich Text File Format

File path

This is an address guide to the location of a file on a disk or network drive. The path tells the computer and user where the file is actually located.

In windows, the back slash (\) is often used to separate parts of the file path. The syntax (arrangement) for writing the file path is **path\filename.file extension.** E.g., E:\Critical data\My Notes\ICT NOTES.docx

Using the file path to identify the file name, extension, number of folders, sub-folders, and route directory

Consider the file path below:

D:\RESOURCES 2025\O LEVEL\S.4 RESOURCES\BOOKS\ ICT-SYLLABUS-compressed.pdf

File Name: ICT-SYLLABUS-compressed (This is the name without the extension.)

File Extension: .pdf

Number of Folders: There are **4 folders** in the path:

- RESOURCES 2025
- LEVEL
- S.4 RESOURCES
- BOOKS

Number of Sub-folders: Assuming "sub-folder" refers to folders nested within another:

- O LEVEL is inside RESOURCES 2025 → sub-folder 1
- S.4 RESOURCES is inside O LEVEL → sub-folder 2
- BOOKS is inside S.4 RESOURCES → sub-folder 3

So, there are **3 sub-folders** (nested levels).

Root Directory: D: (This is the root directory — the main drive where the path starts.)

Measurement of Computer Memory: Units of computer memory

Bit is the smallest unit of data in a computer. A bit is a binary digit represented by either 0 or 1. All data that is in the form of 0s and 1s is binary data or digital data, and is easily manipulated by the computer.

A **byte** is made up of 8 bits of information. One byte is able to store one character, e.g., x, y, 7, 9 in binary form/code. A single space between words also holds one byte.

Unit	**Symbol**	**Bytes (Exact)**	**Approximation**
Kilobyte	KB	1,024 Bytes	~1 Thousand Bytes
Megabyte	MB	1,048,576 Bytes	~1 Million Bytes
Gigabyte	GB	1,073,741,824 Bytes	~1 Billion Bytes
Terabyte	TB	1,099,511,627,776 Bytes	~1 Trillion Bytes
Petabyte	PB	1,125,899,906,842,624 Bytes	~1 Quadrillion Bytes

NOTE: You can get to know the amount of installed memory and other system properties on your PC when you right-click **This PC (**or **Computer)** icon and select **Properties.**

Activity: Converting from one storage measurement to another

1. A computer's hard disk has a storage capacity of 320GB. How many DVDs each of 4.7 GB can be used to provide the same storage for critical data in a business?
2. Namaganda used 12 floppy disks of 1.44 MBs each to back up her essential data for the last ten years. Her fear is that data may soon become unreadable due to outdated floppy disk technology. She now wants to transfer this data onto a flash disk of 1GB. What percentage of space shall be used on the flash disk for this data?
3. Mugisha bought a 32 GB flash drive for use to transfer data from one computer to another. The data to be transferred is 125 GB. How many times will Mugisha copy onto this flash disk to have all the data transferred to another computer?
4. A school's ICT department has received 10 external hard drives, each with a capacity of 500 GB. They plan to back up data stored on school servers totaling 3.5TB. How many of the external hard drives will be needed to store all the data? And what percentage of the hard drives will be unused?
5. A student wants to upload videos totaling 2.1 GB to a cloud service. Each of the videos is approximately 700 MB in size. How many videos can the student upload with the available space?
6. Jane has a 2 TB portable drive. She stores 400 GB of photos, 800 GB of videos, and 300 GB of documents on it. What percentage of the portable drive has she used?
7. A graphics designer stores his project files on DVDs. One of his projects requires 18 GB of storage space. If each DVD can hold 4.7 GB, how many DVDs are needed to store the entire project?
8. An IT technician is backing up files from old 2 GB USB drives. He has 20 such drives, each filled to capacity. He wants to move all the data onto a single external hard drive of 1 TB. Will all the data fit, and how much space will be left?

9. Akello is the school librarian at Greenhill Secondary School. She is responsible for backing up digital records of library inventory, students' borrowing history, and e-books collected over the years. She uses a school desktop computer with a **1 TB hard drive**, but due to increasing data volume and concerns about system failure, she plans to back up the files on external media and cloud storage.

The total data to be backed up includes: Library inventory: **10 GB**, Students' borrowing history: **3.5 GB** and E-books collection: **96.5 GB**

She has access to the following storage devices: DVDs with **4.7 GB** capacity each, USB flash drives with **32 GB** capacity and an online cloud storage account with **150 GB** of free space

Tasks:

(a) Calculate the total size of data Akello needs to back up.

(b) Determine how many DVDs she would need if she chose to back up all the data on DVDs alone.

(c) If she uses the 32 GB flash drives, how many would she need to store all the data?

(d) If she chooses to upload the data to her cloud account, what percentage of the cloud storage will be used?

(e) Suppose Akello backs up the library inventory and student history to DVDs, and the e-books collection to the cloud. Will the cloud storage be enough? Justify your answer.

10. Mr. Ssenoga, the ICT teacher at Hillside College, is preparing digital materials for an inter-school e-learning workshop. He has organized files into three categories:

Student videos: 54 files, each approximately 250 MB

Presentation slides: 35 files, each 5 MB

Software installers: 3 files totaling 4.5 GB

He wants to carry all the materials on a 64 GB flash drive for easy transport. He also considers making DVD backups in case the flash drive fails.

Tasks:
(a) Calculate the total size of the student videos.
(b) Find the total size of all presentation slides.
(c) Determine the total size of all files combined that Mr. Ssenoga wants to carry.
(d) Will all the data fit on the 64 GB flash drive? Show your working.
(e) If Mr. Ssenoga wants to back up the student videos on DVDs of 4.7 GB each, how many DVDs will he need?
(f) What percentage of the flash drive's space will remain **unused** after storing all the files?

Comparing storage devices
Storage devices can be compared in terms of speed of access, portability and storage capacity, among others as summarized in the table below.

Storage Device	**Access Speed**	**Portability**	**Storage Capacity**
Flash disk (USB drive)	Faster access	Very portable	Moderate (4 GB – 1 TB)
Hard Disk (Internal)	Faster access	Bulky	High (500 GB – 10 TB)
Hard Disk (External)	Fast access	Bulky	High (500 GB – 10 TB)
Compact Disc (CD)	Slow	Portable	Low (Up to 700 MB)
Digital Versatile Disc (DVD)	Slow to Moderate	Portable	Low to Moderate (4.7 GB – 8.5 GB)
Magnetic tape	Very Slow	Very Bulky	Very High (Up to several TB)
Memory card (SD/TF)	Moderate	Very portable	Moderate (2 GB – 1 TB)

Burning Data to a Disc
If your computer includes CD/DVD writer, you can use it to copy files and folders to writeable discs. This process is called burning.
Windows operating system come with a preinstalled application that can burn most data files on your computer. If you have to make a disc that will play in music or video player, you should burn using a music or video program.
There are many types of optical discs, among which are:-

(a) CD-R and DVD-R: These discs allow data to be written but the data cannot be physically erased.
(b) CD-RW and DVD-RW: These discs allow data to be written to and erased from them.

Burning or transferring a file to an optical storage
Steps Using Windows Built-in Disc Burning Tool

Step 1: Insert the Disc
Insert a blank CD or DVD into your computer's CD/DVD drive.

Step 2: Choose What to Do
A pop-up should appear with options.
Select "Burn files to disc using File Explorer".
(If not, open File Explorer and click the CD/DVD drive under "This PC.")

Step 3: Choose Disc Format
Enter a disc title (e.g., "Backup_May2025").
Choose one of the following:
Like a USB flash drive – allows files to be added/deleted later (only works with rewritable discs).
With a CD/DVD player – finalizes the disc, suitable for media or sharing.
Click Next.

Step 4: Copy or Drag Files
Open a new window, navigate to the files you want to burn.
Drag and drop or copy and paste them into the CD/DVD window.
Step 5: Start Burning
In the CD/DVD window, click the “Drive Tools” > “Manage” tab, then click “Finish burning”.
Set the disc title (if not done before) and burning speed.
Click Next to start burning.
Step 6: Finalization
Windows will burn the files onto the disc.
Once complete, the disc may eject or show a message saying “The disc was successfully burned.”

Steps to Burn a Disc Using Ashampoo Burning Studio

Step 1: Launch Ashampoo Burning Studio
Open Ashampoo Burning Studio on your computer.

Step 2: Select a Burning Option
Choose "Data Disc" for burning files (documents, photos, videos).
If you're burning music or video files that need to be playable on standalone devices, choose "Audio CD" or "Video DVD" based on your requirements.

Step 3: Select Files to Burn
Click “Add” or drag and drop the files you want to burn into the window.
You can also browse through your folders to select files.

Step 4: Set Disc Name and Preferences
Enter a disc name (e.g., "Backup_May2025").
If desired, you can adjust the burn speed and file system (leave it as default unless you know you need to change it).
ISO9660 is common for compatibility with various devices.
UDF is better for large files and reusability.

Step 5: Insert the Disc
Insert a blank CD/DVD into your optical drive.
Ashampoo should detect the blank disc and show its capacity.

Step 6: Start the Burning Process
Once all files are selected, click the “Burn” button to start the burning process.
Ashampoo will ask for confirmation, then begin burning the files to the disc.
The software will show progress as the data is being written.

Step 7: Eject the Disc
When the process is complete, Ashampoo will notify you, and the disc will be ejected automatically, or you can eject it manually.